

Yonge Li

AI Researcher · AI Engineer · Explainable AI · Machine Learning · Evolutionary Algorithms

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🌐 [LinkedIn](#) 🐙 [GitHub](#) 🌐 [Portfolio](#)

Summary

AI engineer and researcher with an MSc in AI Engineering and 16 years of experience in the CNC machine-tool industry. Specialising in explainable AI, symbolic regression and nature-inspired computing, with a focus on materials science. Seeking a doctoral position in explainable machine learning for scientific discovery.

Education

MSc, Artificial Intelligence Engineering

Aug 2024 – Jun 2026

Jönköping University (JTH), Sweden · 120 ECTS

BSc, Electrical Engineering (Automation)

Sep 2003 – Jun 2007

Heilongjiang University, China

Publications

1. Li Y, Anson E. Interpretable Multi-Property Modeling of Aluminum Alloys Using Symbolic Regression (GP and GE) [Internet] [Dissertation]. 2026. Available from DiVA: [urn:nbn:se:hj:diva-72368](https://nbn-resolving.org/urn:nbn:se:hj:diva-72368)

Thesis Research

Interpretable Multi-Property Modeling of Aluminium Alloys Using Symbolic Regression 2026

Supervisor: Alexandros Tzanetos · In collaboration with COMPtech AB

- Designed a custom BNF grammar with physics-informed operators for Grammatical Evolution (PonyGE2) to predict yield strength and ultimate tensile strength of AA5xxx aluminium alloys.
- Developed a hybrid GE + Ridge fitness function that produces closed-form, interpretable equations rather than black-box predictors.
- Ran 30-seed ablation studies with 5-fold cross-validation across grain-size and data-augmentation configurations to assess stability and generalisation.
- Benchmarked against five baselines (OLS, Ridge, Lasso, Random Forest, XGBoost); compared two symbolic-regression paradigms — Genetic Programming (gplearn) and Grammatical Evolution.
- A journal manuscript based on this work is currently in progress.

Experience

AI Student Researcher — COMPtech AB, Jönköping, Sweden

Oct 2025 – Jun 2026

- Built a PonyGE2-based symbolic-regression system for multi-property prediction of aluminium alloys, from grammar design to evaluation.
- Implemented physics-informed fitness functions and a GE + Ridge hybrid producing interpretable closed-form equations.
- Worked on inverse-design tasks to generate alloy compositions meeting specified target properties; ran experiments on an HPC SLURM cluster.

AI Research Intern — Arfve / layr, Sweden

Jun 2025 – Aug 2025

- Designed the system architecture for a context-aware multimodal AI assistant for wearables (sensor fusion, intent scoring, on-device/edge processing).
- Defined target application scenarios and translated them into design requirements and technical trade-offs.

- 16 years in after-sales technical support and troubleshooting for CNC machine-tool systems (servo drives, spindle units, PLC controllers) and automation equipment.

Selected Projects

- **Hierarchical fashion image classifier (ResNet-50)** — used weighted loss and stratified sampling to lift rare-subclass recall from 0% to over 80%. (*Deep Learning · PyTorch*) [GitHub-Repo](#)
- **ACO for vehicle routing (CVRP)** — tuned ant-colony parameters for 10–26% distance reductions versus Clarke–Wright heuristics. (*Swarm Intelligence*) [GitHub-Repo](#)
- **Agent-based SIR simulation with RL** — Q-learning to optimise interventions, reducing the attack rate from 93% to 45% ($R_0 \approx 10.15$). (*Reinforcement Learning*) [GitHub-Repo](#)
- **Fairness-aware crime prediction** — applied a reweighing technique on a UCI dataset to mitigate algorithmic bias, reaching RMSE 0.1276 and R^2 0.66. (*Machine Learning · Algorithmic Fairness*) [GitHub-Repo](#)

Research Interests

Explainable AI · Materials Science · Symbolic Regression · Evolutionary computation · Optimization

Technical Skills

Languages & Tools: Python · C · Linux · Git · PonyGE2 · gplearn · PyTorch · TensorFlow · scikit-learn · XGBoost · NumPy/Pandas · langchain · LlamaIndex · Ollama · HPC/SLURM · KNIME

Domains & Methods: Interpretable ML · Materials Informatics · Symbolic Regression · Grammatical Evolution · Genetic Programming · Deep Learning · Reinforcement Learning · Cross-validation · Ablation Studies · Feature Selection · Data Augmentation · Retrieval-Augmented Generation (RAG) · Prompt Engineering

References

Dr. Alexandros Tzanetos — Assistant Professor, Jönköping University (JTH), Sweden
alexandros.tzanetos@ju.se — MSc thesis supervisor

Dr. Vladimir Tarasov — Associate Professor, Jönköping University (JTH), Sweden
vladimir.tarasov@ju.se — MSc thesis examiner